

- 74** An investigation was carried out to study the effect of protease concentration on the rate of breakdown of a protein into amino acids.

A protease solution of known concentration was diluted and used in each experiment, and all other variables were kept constant. The table shows the results.

<i>percentage concentration of protease</i>	10	20	30	40	50	60	70
<i>rate of reaction / arbitrary units</i>	1.8	3.5	5.3	6.0	6.3	6.3	6.3

Which of the following statements is/are correct about this investigation?

- 1** Between concentrations of 10% and 40%, the rate of reaction has increased as more active sites are available.
 - 2** Between concentrations of 50% and 70%, the enzyme concentration is the limiting factor.
 - 3** At a protease concentration of 60%, the rate of reaction will be faster if substrate concentration is increased.
- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3